HUNARMAND PUNJAB PROGRAM

Batch-3 Final Project

National Cyber Security

TOTAL MARKS	40	LONG QUESTIONS	2	MARKS EACH	20

Q1 Authorized Network Reconnaissance & Traffic Analysis Lab — 20 MARKS

Scenario: Build an isolated cyber-security lab using Windows 7 and a Linux/Kali machine. Assume both systems belong to your own training environment. Perform network discovery, information gathering and packet analysis, then document your findings professionally.

Complete the following practical work:

- Lab Setup: Show the practical environment, IP addresses and connectivity between the Windows 7 and Linux/Kali systems.
- CLI & Linux Commands: Demonstrate useful Windows command-line and Linux commands for checking IP configuration, connectivity, files/directories and basic network information.
- Network Concepts: Briefly identify the relevant network model/layers and explain how communication takes place between the two lab machines.
- Network Discovery: Use Nmap on the authorized Windows 7 lab target and also demonstrate either Zenmap or Angry IP Scanner. Record discovered host information, open ports and detected services.
- Information Gathering: Perform basic and advanced information gathering on the lab target. If search operators/dorks are demonstrated, use only an instructor-approved or owned test domain and do not target unrelated public systems.
- Wireshark Analysis: Capture traffic generated inside the lab and identify at least three protocol examples such as ARP, ICMP, DNS, TCP or HTTP. Apply suitable display filters and explain what the packets show.
- Vulnerability Identification: Based on scan results, identify potential weaknesses or outdated services and explain why they may create risk. Do not attack any system outside the lab.
- Evidence: Attach clear screenshots of lab connectivity, CLI commands, scan results, Wireshark filters/packets and your findings.

Final Submission: Submit one PDF report containing screenshots, findings and a short conclusion with recommended security improvements.

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Q2 Web Application Pentesting, Vulnerability Validation & Security Report — 20 MARKS

Scenario: Use DVWA and the intentionally vulnerable Windows 7 machine inside your local training lab. Perform a controlled web application security assessment, validate taught vulnerabilities and then demonstrate protection/remediation.

Complete the following practical work:

- DVWA + OWASP: Set up/open DVWA and map the vulnerabilities you test to relevant OWASP Top 10 concepts covered in the course.
- SQL Injection: Demonstrate SQL Injection only against DVWA in the authorized lab. Capture evidence showing the vulnerable behavior and explain the security impact in simple words.
- Windows 7 Vulnerability Validation: Use Nmap/Zenmap findings to check whether the intentionally vulnerable Windows 7 lab machine exposes the service associated with the EternalBlue/MS17-010 lesson. If your trainer provided a controlled exploitation exercise, include only proof of the lab result and the affected service - do not include reusable exploit commands or attack instructions.
- Firewall Protection: Apply a firewall rule or protection step taught in the course to reduce access to DVWA or another exposed lab service. Show before/after evidence.
- Security Verification: Re-test after the protection step and explain what changed.
- Digital Forensics Overview: Identify what evidence you would preserve after the test, such as timestamps, screenshots, IP addresses, logs, packet captures or affected files.
- Professional Reporting: Prepare findings in a cyber-security report format including Title, Scope, Evidence, Vulnerability, Risk, Impact, Recommendation and Final Conclusion.
- Evidence: Attach clear screenshots of DVWA, OWASP-related testing, SQL Injection result, Windows 7 vulnerability check, firewall rule and re-test result.

Important: All scanning, testing and vulnerability validation must remain inside your own/instructor-approved lab environment.